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Riesz Learning for Causal Mediation Analysis

Richard Liu, Nicholas T Williams, Kara E. Rudolph, and Iván Díaz

American Causal Inference Conference (ACIC) 2026



- All content comes from this work:
  Liu R, Williams NT, Rudolph KE, Díaz I. General targeted machine learning for modern causal mediation analysis. arXiv preprint arXiv:2408.14620. 2024 Aug 26.
  But we note that this talk only covers the parts related to Riesz learning.

[1] Williams NT, Hines OJ, Rudolph KE. Riesz representers for the rest of us. arXiv preprint arXiv:2507.19413. 2025 Jul 25.

[2] Balkus SV, Testa C, Hejazi NS. A Riesz Representer Perspective on Targeted Learning. arXiv preprint arXiv:2604.21721. 2026 Apr 23.

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  But we note that this talk only covers the parts related to Riesz learning.
- It is very possible that you have seen a lot of similar discussion in other papers[1][2]. I would not claim that this talk is "new" to you due to this reason.

[1] Williams NT, Hines OJ, Rudolph KE. Riesz representers for the rest of us. arXiv preprint arXiv:2507.19413. 2025 Jul 25.

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Acknowledgement

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Nicholas T Williams



Kara E. Rudolph



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Introduction



- *Riesz learning*[3][4][5] has been widely applied in estimating many causal effects.
  - This talk focuses on examples in *causal mediation analysis*.

[3] Chernozhukov V, Newey WK, Singh R. Automatic debiased machine learning of causal and structural effects. Econometrica. 2022 May;90(3):967-1027.

[4] Chernozhukov V, Newey W, Quintas-Martınez VM, Syrgkanis V. Riesznet and forestriesz: Automatic debiased machine learning with neural nets and random forests. InInternational Conference on Machine Learning 2022 Jun 28 (pp. 3901-3914). PMLR.

[5] Chernozhukov V, Newey WK, Singh R. Debiased machine learning of global and local parameters using regularized Riesz representers. The Econometrics Journal. 2022 Sep;25(3):576-601.

- *Riesz learning*[3][4][5] has been widely applied in estimating many causal effects.
  - This talk focuses on examples in *causal mediation analysis*.
- An important advantage is to estimate Riesz representers (RRs) **directly** without knowing the analytical form.
  - E.g. Inverse propensity score ($\frac{I(A=1)}{P(A=1|W)}$).

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  - This talk focuses on examples in *causal mediation analysis*.
- An important advantage is to estimate Riesz representers (RRs) **directly** without knowing the analytical form.
  - E.g. Inverse propensity score ($\frac{I(A=1)}{P(A=1|W)}$).
- This will help resolving the estimation problem when the RR is difficult to estimate.

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What I will cover

- Riesz learning for a nested conditional expectation/series of sequential regressions.
- Examples in causal mediation analysis.

What I will not cover

- An introduction to semiparametric theories.
- How to do sequential regression.
- Definitions, interpretations and identification assumptions of mediation parameters discussed in this talk.

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Definition (Von Mises expansion)

Let $\psi : \mathcal{P} \to R$ be a functional from the non-parametric model to the real line. The functional admits a first-order von Mises expansion if it satisfies:

$$\psi(F) - \psi(P) = -E_P[\varphi(X; F)] + R_2(F, P)$$

for any $F, P \in \mathcal{P}$, where $\varphi(x; F)$ is a function satisfying $E_F[\varphi(X; \psi(F))] = 0$ and $E_F[\varphi^2(X; \psi(F))] < \infty$, and the term $R_2(F, P)$ is a second-order remainder term depending on products or squares of differences between F and P. The function φ is typically referred to as a gradient of the first-order expansion.

Takeaway: Riesz learning solves an estimation problem in $\varphi(X; F)$, the (centered) *efficient influence function* (EIF) .

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An Overview of Sequential Regression and Its Efficient Influence Function



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$$\int_w E(Y \mid A = 1, w) \mathrm{d}P(w) = E[E(Y \mid A = 1, W)]$$

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$$\int_w E(Y \mid A = 1, w) \mathrm{d}P(w) = E[E(Y \mid A = 1, W)]$$

Example 2 (Natural Direct/Indirect Effects, $O = (W, A, M, Y)$)

$$\int_w \int_m E(Y \mid A = a_1, m, w) \mathrm{d}P(m \mid A = a_2, w) \mathrm{d}P(w)$$
$$= E\{E[E(Y \mid A = a_1, M, W) \mid A = a_2, W]\}$$

Example 3 (Longitudinal Treatment Effects, $O = (L_1, A_1, L_2, A_2, Y)$)

$$\int_{a_1, l_1} \int_{a_2, l_2} E(Y \mid \bar{a}_2, \bar{l}_2) \mathrm{d}P(a_2, l_2 \mid a_1, l_1) \mathrm{d}P(a_1, l_1)$$
$$= E\{E[E(Y \mid \bar{a}_2, \bar{l}_2) \mid a_1, l_1]\}$$

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$$= E\{E[E(Y \mid \bar{a}_2, \bar{l}_2) \mid a_1, l_1]\}$$

All such statistical functionals can be easily estimated by **running a sequence of regressions**.

Conjecture: For such sequential regression functionals, EIF can normally be written as a summation of products, with each one being a product of **a Riesz Representer** and **an outcome residual**.

Their EIFs look like...

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Why this is a conjecture?

For every example discussed in the talk, the claim above is true. Though I guess this works for every sequential regression, a formal analysis is still missing to the best of my knowledge.

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Example 1: Consider $\psi = E[E(Y \mid A = 1, W)]$, then,

$$\varphi = \frac{I(A=1)}{P(A=1 \mid W)}[Y - E(Y \mid A = 1, W)] + 1 \cdot [E(Y \mid A = 1, W) - \psi]$$

Example 2: Consider $\psi = E\{E[\underbrace{E(Y \mid A = a_1, M, W)}_{Q(M,W)} \mid A = a_2, W]\}$,

then,

$$\varphi = \left[\frac{I(A = a_1)}{P(A = a_1 \mid W)} \frac{P(M \mid A = a_2, W)}{P(M \mid A = a_1, W)}\right][Y - E(Y \mid a_1, M, W)]$$
$$+ \frac{I(A = a_2)}{P(A = a_2 \mid W)}\{Q(M, W) - E[Q(M, W) \mid a_2, W]\}$$
$$+ 1 \cdot \{E[Q(M, W) \mid a_2, W] - \psi\}$$

Theorem 1 (Sequential Riesz EIF, (Balkus et al., 2026)). *Consider the bounded linear functional* $\Psi(P) = E_P\big[h_1(\bar{A}_1, \bar{L}_1; \bar{Q}_1)\big]$, *where* $\bar{Q}_t$ *is a bounded linear functional defined sequentially such that, for* $t = 1, \ldots, T$, *we have* $\bar{Q}_t(\bar{A}_t, \bar{L}_t) = E_P\big[h_{t+1}(\bar{A}_{t+1}, \bar{L}_{t+1}; \bar{Q}_{t+1}) \,\big|\, \bar{A}_t, \bar{L}_t\big]$, *with* $\bar{Q}_T(\bar{A}_T, \bar{L}_T) := E_P\big(Y \mid \bar{A}_T, \bar{L}_T\big)$. *Let* α_t *denote the Riesz representer for* $\bar{Q}_{t+1}$ *in the functional* $E_P\big[h_{t+1}(\bar{A}_{t+1}, \bar{L}_{t+1}; \bar{Q}_{t+1}) \,\big|\, \bar{A}_t, \bar{L}_t\big]$. *Then, defining the outcome* $h_T(\bar{A}_{T+1}, \bar{L}_{T+1}; \bar{Q}_{T+1}) := Y$, *the EIF of* $\Psi(P)$ *is*

$$\phi(P)(O) = \sum_{t=1}^{T} \prod_{k=1}^{t} \alpha_k(\bar{A}_k, \bar{L}_k)\big[h_{t+1}(\bar{A}_{t+1}, \bar{L}_{t+1}; \bar{Q}_{t+1}) - \bar{Q}_t(\bar{A}_t, \bar{L}_t)\big] + 1 \times [h_1(\bar{A}_1, \bar{L}_1; \bar{Q}_1) - \psi].$$

[6] Balkus SV, Testa C, Hejazi NS. A Riesz Representer Perspective on Targeted Learning. arXiv preprint arXiv:2604.21721. 2026 Apr 23.

- Susmann H, Williams NT, Rudolph KE, Díaz I. Longitudinal generalizations of the average treatment effect on the treated for multi-valued and continuous treatments. arXiv preprint arXiv:2405.06135. 2024 May 9.
- Some earlier ones should exist...

The estimation of the EIF of a sequential regression can be decomposed into two tasks:

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The estimation of the EIF of a sequential regression can be decomposed into two tasks:

- Estimating **a series of outcome residuals**, which is easy.
- Estimating **a series of RRs**, which can be very hard without Riesz learning.

Objective: Provide a **nonparametric estimation toolkit** of the following mediation parameters:

- Natural effects
- Decision theoretic effects
- Randomized effects
- Organic effects
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even when mediators are continuous and/or high-dimensional.

Challenge 1: Not all identification formulas can be written as sequential regressions.

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Example (Randomized effects): $O = (W, A, Z, M, Y)$

$$\psi = \int_w \int_m \int_z E(Y \mid A = a_1, z, m, w)\, \mathrm{d}P(m \mid A = a_2, w)\, \mathrm{d}P(z \mid A = a_1, w)\, \mathrm{d}P(w)$$

$$= E\left[E\left\{ \underbrace{\int_z E(Y \mid A = a_1, z, M, W)\, \mathrm{d}P(z \mid A = a_1, W)}_{\neq E[E(Y|A=a_1,Z,M,W)|A=a_1,W]} \,\middle|\, A = a_2, W \right\} \right]$$

Therefore, $\int_z E(Y \mid A = a_1, z, M, W)\, \mathrm{d}P(z \mid A = a_1, W)$ needs to be evaluated first before running sequential regressions (can be very hard when $Z = z$ is continuous and/or high-dimensional).

Challenge 2: Estimation of the RRs is elusive with continuous and/or high-dimensional mediators.

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Example: Consider $\psi = E\{E[\underbrace{E(Y \mid A = a_1, M, W)}_{Q(M,W)} \mid A = a_2, W]\}$,

then,

$$\varphi = \left[\frac{I(A = a_1)}{P(A = a_1 \mid W)} \frac{P(M \mid A = a_2, W)}{P(M \mid A = a_1, W)}\right][Y - E(Y \mid a_1, M, W)]$$
$$+ \frac{I(A = a_2)}{P(A = a_2 \mid W)}\{Q(M, W) - E[Q(M, W) \mid a_2, W]\}$$
$$+ 1 \cdot \{E[Q(M, W) \mid a_2, W] - \psi\}$$

Bayes Re-parametrization?

- Someone may argue that estimation is not a problem even when the data type is tricky.
- For example, $\frac{P(M|A=0,W)}{P(M|A=1,W)} = \frac{P(A=0|M,W)}{P(A=1|M,W)} \frac{P(A=1|W)}{P(A=0|W)}$.

[7] Rudolph KE, Williams NT, Diaz I. Practical causal mediation analysis: extending nonparametric estimators to accommodate multiple mediators and multiple intermediate confounders. Biostatistics. 2024 Oct;25(4):997-1014.

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- While such trick works for natural effects, a recent paper[7] found that, at least for randomized effects (actually, also for recanting twin effects), the assumptions for the parameters to have a good statistical property (e.g. asymptotic normality) are more difficult to satisfy.

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  - E.g. $\frac{I(A{=}1)}{P(A|W)} \frac{P(M|A{=}a_2,Z,W)}{P(M|A,Z,W)} \frac{P(Z|A{=}a_3,W)}{P(Z|A,W)}$.

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- While such trick works for natural effects, a recent paper[7] found that, at least for randomized effects (actually, also for recanting twin effects), the assumptions for the parameters to have a good statistical property (e.g. asymptotic normality) are more difficult to satisfy.
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- Therefore, our objective cannot be fulfilled by using a simple reparametrization trick.

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Estimating RRs from a mediation statistical functional using Riesz Learning



- **Idea of Riesz learning**: For an arbitrary b, if $E[b(X, \theta)]$ is a continuous linear functional of θ, then by the Riesz representation theorem there exists a unique function α such that $E[b(X; \theta)] = E[\alpha(X)\theta(X)]$ for all θ with $E[\theta(X)^2] < \infty$.

- **Idea of Riesz learning**: For an arbitrary b, if $E[b(X,\theta)]$ is a continuous linear functional of θ, then by the Riesz representation theorem there exists a unique function α such that $E[b(X;\theta)] = E[\alpha(X)\theta(X)]$ for all θ with $E[\theta(X)^2] < \infty$.
- Starting from the MSE, we can estimate α by the following way:

$$\begin{aligned}
\alpha &= \arg\min_{\tilde{\alpha}} E[(\alpha(X) - \tilde{\alpha}(X))^2] \\
&= \arg\min_{\tilde{\alpha}} E[\alpha(X)^2 - 2\tilde{\alpha}(X)\alpha(X) + \tilde{\alpha}(X)^2] \\
&= \arg\min_{\tilde{\alpha}} E[\tilde{\alpha}(X)^2 - 2\tilde{\alpha}(X)\alpha(X)] \\
&= \arg\min_{\tilde{\alpha}} E[\tilde{\alpha}(X)^2 - 2b(X,\tilde{\alpha})].
\end{aligned}$$

- We do not need to assume a form of α, due to which we do not need to do density estimation.

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Riesz Learning Steps

1. Cleverly define bs. For a nested conditional expectation, there could be multiple bs.
2. Find corresponding αs.
3. Estimate αs by Riesz learning.

Example 1: Consider $\psi = E[E(Y \mid A = 1, W)]$.

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We define $\theta(A, W) = E(Y \mid A, W)$, and $b(A, W; \theta) = \theta(1, W)$, then,

$$\psi = E[b(A, W; \theta)] = E[\alpha(A, W)\theta(A, W)].$$

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$$\psi = E[b(A, W; \theta)] = E[\alpha(A, W)\theta(A, W)].$$

By the change-of-measure in deriving an IPW estimator, we have

$$\psi = E[\frac{I(A=1)}{P(A=1 \mid W)} E(Y \mid A, W)],$$

meaning that $\alpha_1(A, W) = \frac{I(A=1)}{P(A=1|W)}$ is the RR for $b(A, W; \theta)$.

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meaning that $\alpha_1(A, W) = \frac{I(A=1)}{P(A=1|W)}$ is the RR for $b(A, W; \theta)$. At last, we can estimate such α_1 by solving

$$\alpha_1 = \arg\min_{\tilde{\alpha}} E[\tilde{\alpha}(A, W)^2 - 2\tilde{\alpha}(1, W)]$$

Example 2: Consider $\psi = E\{E[E(Y \mid A = a_1, M, W) \mid A = a_2, W]\}$.

Example 2: Consider $\psi = E\{\overbrace{E[\underbrace{E(Y \mid A = a_1, M, W)}_{\theta_2(a_1,M,W)} \mid A = a_2, W]}^{\theta_1(a_2,W)}\}$.

Define $b_1(A, W; \theta_1) = \theta_1(a_2, W)$, $b_2(A, M, W; \theta_2) = \theta_2(a_1, M, W)$, and we have

$$E(b_1) = E(\alpha_1\theta_1), E[E(b_2 \mid A = a_2, W)] = E(\alpha_2\theta_2).$$

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$$E(b_1) = E(\alpha_1\theta_1), E[E(b_2 \mid A = a_2, W)] = E(\alpha_2\theta_2).$$

Then, α_1 can be estimated by solving

$$\alpha_1 = \arg\min_{\tilde{\alpha}} E[\tilde{\alpha}(A, W)^2 - 2\tilde{\alpha}(a_2, W)],$$

For α_2, we have

$$\alpha_2 = \arg\min_{\tilde{\alpha}} E\left\{\tilde{\alpha}(A, M, W)^2 - 2E[b_2(A, M, W; \tilde{\alpha}) \mid A = a_2, W]\right\},$$

For α_2, we have

$$\alpha_2 = \arg\min_{\tilde{\alpha}} E\left\{\tilde{\alpha}(A,M,W)^2 - 2\underbrace{E[b_2(A,M,W;\tilde{\alpha}) \mid A=a_2, W]}_{q_1(a_2,W)}\right\},$$

because

$$\begin{aligned} &E[q_1(a_2,W)] = E[b_1(A,W;q_1)] = E[\alpha_1(A,W)q_1(A,W)] \\ &= E\left\{E[\alpha_1(A,W)b_2(A,M,W;\tilde{\alpha}) \mid A=a_2, W]\right\} \\ &= E[\alpha_1(A,W)b_2(A,M,W;\tilde{\alpha})], \end{aligned}$$

For α_2, we have

$$\alpha_2 = \arg\min_{\tilde{\alpha}} E\left\{\tilde{\alpha}(A,M,W)^2 - 2\underbrace{E[b_2(A,M,W;\tilde{\alpha}) \mid A=a_2, W]}_{q_1(a_2,W)}\right\},$$

because

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we have

$$\alpha_2 = \arg\min_{\tilde{\alpha}} E[\tilde{\alpha}(A,M,W)^2 - 2\alpha_1(A,W)b_2(A,M,W;\tilde{\alpha})],$$

In fact, one can prove that

$$\alpha_1(A, W) = \frac{I(A = a_2)}{P(A = a_2 \mid W)}$$

$$\alpha_2(A, M, W) = \frac{I(A = a_1)}{P(A = a_1 \mid W)} \frac{P(M \mid A = a_2, W)}{P(M \mid A = a_1, W)}$$

Recall

$$\begin{aligned}\varphi = &\left[\frac{I(A = a_1)}{P(A = a_1 \mid W)} \frac{P(M \mid A = a_2, W)}{P(M \mid A = a_1, W)}\right][Y - E(Y \mid a_1, M, W)] \\ &+ \frac{I(A = a_2)}{P(A = a_2 \mid W)}\{Q(M, W) - E[Q(M, W) \mid a_2, W]\} \\ &+ 1 \cdot \{E[Q(M, W) \mid a_2, W] - \psi\}\end{aligned}$$

Riesz learning for a sequential regression suggests a sequential procedure as well.

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Quick Summary of Our Contribution



We introduced an auxiliary random variable Z^π such that

$$Z^\pi \mid (A, W) \sim Z \mid (A, W),$$
$$Z^\pi \perp M \mid (A, W),$$

so that we can transform a multivariate integral into a sequential regression.

- $A \in \{0, 1\}$: a binary treatment or exposure variable (general case in the latter part).
- Y: continuous/binary outcome.
- M: a vector of mediators.
- Z: a vector of intermediate confounders.
- W: a vector of observed covariates.
- The observed data for unit i can be augmented as $(W_i, A_i, Z_i, Z_i^\pi, M_i, Y_i)$.

Defitinion 1

For values $(a_1, \ldots, a_4) \in \{0,1\}^4$, we define

$$\psi^R(a_1,a_2,a_3,a_4) = E\Big[E\Big[E\big[E(Y \mid A=a_1, Z^\pi, M, W) \,\big|\, A=a_2, M, W\big] \,\Big|\, A=a_3, Z, W\Big] \,\Big|\, A=a_4, W\Big]$$

$$= \int_w \int_z \int_m \int_{z'} E(Y \mid a_1, Z^\pi = z', m, w) \mathrm{d}P(z' \mid a_2, m, w) \mathrm{d}P(m \mid a_3, Z=z, w) \mathrm{d}P(z \mid a_4, w) \mathrm{d}P(w)$$

as the *randomized parameter* and

$$\psi^N(a_1,a_2,a_3) = E\Big[E\Big[E\big[E(Y \mid A=a_1, Z, M, W) \,\big|\, A=a_2, Z, W\big] \,\Big|\, A=a_3, W\Big]\Big],$$

$$= \int_w \int_z \int_m E(Y \mid A=a_1, z, m, w) \mathrm{d}P(m \mid A=a_2, z, w) \mathrm{d}P(z \mid A=a_3, w) \mathrm{d}P(w)$$

as the *natural parameter*.

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Proposition 1

The identification formulas mentioned above can be written in terms of ψ^N and ψ^R as:

$$\text{NDE} = \psi^N(1,0,0) - \psi^N(0,0,0), \qquad \text{NIE} = \psi^N(1,1,1) - \psi^N(1,0,0),$$
$$\text{RIDE} = \psi^R(1,1,0,0) - \psi^R(0,0,0,0), \qquad \text{RIIE} = \psi^R(1,1,1,1) - \psi^R(1,1,0,0),$$

as well as

$$\phi(P_1) = \psi^N(1,1,1) - \psi^N(0,1,1), \qquad \phi(P_2) = \psi^R(0,1,1,1) - \psi^R(0,0,1,1),$$
$$\phi(P_3) = \psi^R(0,0,1,1) - \psi^R(0,0,1,0), \qquad \phi(P_4) = \psi^N(0,1,0) - \psi^N(0,0,0).$$

- The success in unifying all these common and important causal mediation parameters allows the package `crumble` to be general.

Theorem 1

The functions

$$\begin{aligned}\varphi^N(X;\eta^N) &= \bar{\varphi}^N(X;\eta^N) - \psi^N \\ &= \alpha_3^N(A,Z,M,Y)\{Y - \theta_3^N(A,Z,M,W)\} \\ &+ \alpha_2^N(A,Z,W)\{b_3^N(Z,M,W;\theta_3^N) - \theta_2^N(A,Z,W)\} \\ &+ \alpha_1^N(A,W)\{b_2^N(Z,W;\theta_2^N) - \theta_1^N(A,W)\} \\ &+ b_1^N(W;\theta_1^N) - \psi^N,\end{aligned}$$

and

$$\begin{aligned}\varphi^R(X;\eta^R) &= \bar{\varphi}^R(X;\eta^R) - \psi^R \\ &= \alpha_4^R(A,Z,M,W)\{Y - \theta_4^R(A,Z,M,W)\} \\ &+ \alpha_3^R(A,M,W)\{b_4^R(Z^\pi,M,W;\theta_4^R) - \theta_3^R(A,M,W)\} \\ &+ \alpha_2^R(A,Z,W)\{b_3^R(M,W;\theta_3^R) - \theta_2^R(A,Z,M)\} \\ &+ \alpha_1^R(A,W)\{b_2^R(Z,W;\theta_2^R) - \theta_1^R(A,W)\} \\ &+ b_1^R(W;\theta_1^R) - \psi^R\end{aligned}$$

satisfy the first order von Mises expansion for the parameters ψ^N and ψ^R, respectively.

For ψ^N, we define

$$\begin{aligned}
\theta_3^N(a,z,m,w) &= E(Y \mid A=a, Z=z, M=m, W=w), & b_3^N(z,m,w;\theta_3^N) &= \theta_3^N(a_1,z,m,w),\\
\theta_2^N(a,z,w) &= E[b_3^N(Z,M,W;\theta_3^N) \mid A=a, Z=z, W=w], & b_2^N(z,w;\theta_2^N) &= \theta_2^N(a_2,z,w),\\
\theta_1^N(a,w) &= E[b_2^N(Z,W;\theta_2^N) \mid A=a, W=w], & b_1^N(w;\theta_1^N) &= \theta_1^N(a_3,w),
\end{aligned} \tag{1}$$

as well as

$$\begin{aligned}
\alpha_1^N(a,w) &= \frac{I(A=a_3)}{P(a \mid w)},\\
\alpha_2^N(a,z,w) &= \frac{I(A=a_2)}{P(a \mid w)}\frac{P(z \mid A=a_3,w)}{P(z \mid a,w)},\\
\alpha_3^N(a,z,m,w) &= \frac{I(A=a_1)}{P(a \mid w)}\frac{P(m \mid A=a_2,z,w)}{P(m \mid a,z,w)}\frac{P(z \mid A=a_3,w)}{P(z \mid a,w)}.
\end{aligned} \tag{2}$$

- The following lemma shows that α_j^N corresponds to the "unique function α" in Riesz learning.

Lemma 1(ψ^N)

If additionally, we define

$$\begin{aligned}
&\Psi_0^N(\theta_0^N) = \theta_0^N \\
&\Psi_1^N(\theta_1^N) = E\{b_1^N(W;\theta_1^N)\}; \\
&\Psi_2^N(\theta_2^N) = E[E\{b_2^N(Z,W;\theta_2^N) \mid A = a_3, W\}]; \\
&\Psi_3^N(\theta_3^N) = E\big[E\big\{E[b_3^N(Z,M,W;\theta_3^N) \mid A = a_2, Z, W] \mid A = a_3, W\big\}\big].
\end{aligned} \tag{3}$$

Then

$$\Psi_j^N(\theta_j^N) = E[\alpha_j^N(X)\theta_j^N(X)]$$

for all θ_j^N.

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Proposition 2

The MSE loss functions for $(\alpha_1^N, \alpha_2^N, \alpha_3^N)$ may be expressed as

$$\begin{aligned}
\alpha_1^N &= \arg\min_{\alpha} E[\alpha(A,W)^2 - 2b_1^N(W,\alpha)], \\
\alpha_2^N &= \arg\min_{\alpha} E\{\alpha(A,Z,W)^2 - 2\alpha_1^N(A,W)b_2^N(Z,W,\alpha)\}, \\
\alpha_3^N &= \arg\min_{\alpha} E\{\alpha(A,Z,M,W)^2 - 2\alpha_2^N(A,Z,W)b_3^N(Z,M,W,\alpha)\},
\end{aligned} \tag{4}$$

and the MSE loss functions for $(\alpha_1^R, \alpha_2^R, \alpha_3^R, \alpha_4^R)$ may be expressed as

$$\begin{aligned}
\alpha_1^R &= \arg\min_{\alpha} E[\alpha(A,W)^2 - 2b_1^R(W,\alpha)], \\
\alpha_2^R &= \arg\min_{\alpha} E\{\alpha(A,Z,W)^2 - 2\alpha_1^R(A,W)b_2^R(Z,W,\alpha)\}, \\
\alpha_3^R &= \arg\min_{\alpha} E\{\alpha(A,M,W)^2 - 2\alpha_2^R(A,Z,W)b_3^R(M,W;\alpha)\}, \\
\alpha_4^R &= \arg\min_{\alpha} E\{\alpha(A,Z,M,W)^2 - 2\alpha_3^R(A,M,W)b_4^R(Z^\pi,M,W;\alpha)\}.
\end{aligned} \tag{5}$$

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Simulation



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$$
\begin{aligned}
W = (W_1, W_2, W_3)^\top &\sim Be(2, 3) \\
A|W &\sim Bern(\text{logit}^{-1}(0.5W_1 + 0.5W_2 - 1)) \\
Z_1|A, W &\sim TN(-1, 1, -0.4 + \epsilon A + 0.2W_3^2, 1) \\
Z_2|Z_1, A, W &\sim TN(-1, 1, 0.2 - \epsilon A + 0.5\sin(W_2), 1) \\
M_1|Z_1, Z_2, A, W &\sim TN(-1, 1, -0.5 + \lambda_1 Z_1 + \lambda_2 A + 0.4W_2 + 0.2W_3, 1) \\
M_2|M_1, Z_1, Z_2, A, W &\sim TN(-1, 1, -0.5 + \lambda_1 Z_2 + \lambda_2 A + 0.4W_1 + 0.2W_3, 1) \\
Y|M_1, M_2, Z_1, Z_2, A, W &\sim N(0.2M_1 + 0.2M_2 + \gamma_1 Z_1/2 + \gamma_1 Z_2/2 + \gamma_2 A - 0.5\cos(W_1) - 1.5, 1).
\end{aligned}
$$

Here, $(\epsilon, \lambda_1, \lambda_2, \gamma_1, \gamma_2)^\top = (0.5, 0.4, 0.6, 0.6, 0.4)^\top$ are pre-specified parameters; $Be(\alpha, \beta)$ denotes a Beta distribution with shape parameters α and β; $Bern(\pi)$ denotes a Bernoulli distribution with probability π; $N(\mu, \sigma^2)$ denotes a normal distribution with mean μ and variance σ^2; $TN(a, b, \mu, \sigma^2)$ denotes a truncated normal distribution derived from $N(\mu, \sigma^2)$, but with values in $[a, b]$.

Table 2: Simulation results for our method.

		recanting twins					interventional		natural	
n	metrics	AY	AZY	AZMY	AMY	Int. Confounder	IDE	IIE	NDE	NIE
500	Bias	0.016	-0.014	-0.020	0.004	-0.003	-0.005	0.005	0.000	-0.005
	$\sqrt{n}$-Bias	0.386	-0.302	-0.439	0.092	-0.058	-0.102	0.115	0.000	-0.121
	nMSE	9.009	15.79	3.500	4.917	9.948	5.135	11.53	7.189	3.143
	Coverage	0.942	0.940	0.952	0.950	0.960	0.942	0.960	0.954	0.940
1000	Bias	0.023	-0.033	-0.015	0.000	0.003	-0.005	0.002	0.005	-0.004
	$\sqrt{n}$-Bias	0.725	-1.032	-0.464	-0.002	0.103	-0.158	0.052	0.166	-0.128
	nMSE	8.961	10.82	2.333	3.270	5.722	4.416	10.03	5.996	1.306
	Coverage	0.932	0.930	0.942	0.946	0.970	0.950	0.948	0.962	0.942
2000	Bias	0.021	-0.026	-0.014	-0.002	0.005	0.000	0.000	0.003	0.000
	$\sqrt{n}$-Bias	0.951	-1.155	-0.628	-0.077	0.240	0.014	0.016	0.142	0.007
	nMSE	7.709	4.760	1.513	2.276	4.670	3.597	10.123	5.972	1.029
	Coverage	0.948	0.888	0.934	0.958	0.964	0.936	0.940	0.956	0.966

- We unified the identification results of five commonly-seen mediation analysis methods into two statistical estimands.
- (Not shown in the slides) Our estimators for these two estimands have double robustness, asymptotic linearity, and weak convergence.
- With the aid of Riesz learning, our estimators can achieve optimal efficiency even when Z, M are continuous and/or multivariate.
- Package `crumble` (on CRAN) was provided for unified mediation analysis.
- Paper link: https://arxiv.org/abs/2408.14620



THANK YOU

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Theories, Construction of Z^{π}



- We use cross-fitting to avoid a Donsker assumption.
- Let $\mathcal{V}_1, \ldots, \mathcal{V}_J$ denote a random partition of the index set $\{1, \ldots, n\}$ into J prediction sets of approximately the same size.
- In addition, for each j, the associated training sample is given by $\mathcal{T}_j = \{1, \ldots, n\} \setminus \mathcal{V}_j$. Let $\hat{\eta}_j^N$ and $\hat{\eta}_j^R$ denote the parameter estimates using training data $\mathcal{T}_j$.
- Let $j(i)$ denote the index of the prediction set which contains observation i.

The final cross-fitted estimators of ψ^N and ψ^R are computed as:

$$\hat{\psi}^N = \frac{1}{n}\sum_{i=1}^{n} \bar{\varphi}^N(X_i, \hat{\eta}^N_{j(i)}), \quad \text{and} \quad \hat{\psi}^R = \frac{1}{n}\sum_{i=1}^{n} \bar{\varphi}^R(X_i, \hat{\eta}^R_{j(i)}), \tag{6}$$

respectively.

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Assumption 1

For $E \in \{N, R\}$, assume the estimated functions $\hat{\alpha}_j^E$ are bounded in the sense that there exists a constant $M < \infty$ such that $P(\hat{\alpha}_j^E(X) \leq M) = P(\alpha_j^E(X) \leq M) = 1$,

For Assumption 2, we define $\tilde{\theta}^N$ as

$$\tilde{\theta}^N_{2,j(i)}(a, z, w) = E[b_3^N(Z, M, W; \hat{\theta}^N_{3,j(i)}) \mid A = a, Z = z, W = w];$$
$$\tilde{\theta}^N_{1,j(i)}(a, w) = E[b_2^N(Z, W; \hat{\theta}^N_{2,j(i)}) \mid A = a, W = w],$$

Assumption 2

For $E \in \{N, R\}$, assume $R_2^E(\hat{\eta}^E_{j(i)}, \eta^E) = o_P(n^{-1/2})$, where

$R_2^N(\hat{\eta}^N_{j(i)}, \eta^N) = \sum_{k=1}^3 E\left[\{\hat{\alpha}^N_{k,j(i)}(X) - \alpha_k^N(X)\}\{\hat{\theta}^N_{k,j(i)}(X) - \tilde{\theta}_k^N(X)\}\right]$,

$R_2^R(\hat{\eta}^R_{j(i)}, \eta^R) = \sum_{k=1}^4 E\left[\{\hat{\alpha}^R_{k,j(i)}(X) - \alpha_k^R(X)\}\{\hat{\theta}^R_{k,j(i)}(X) - \tilde{\theta}_k^R(X)\}\right]$.

Theorem 2

Assume Assumption 1 holds, then we have

$$\hat{\psi}^E - \psi^E = \frac{1}{n}\sum_{i=1}^{n}\{\varphi^E(X_i;\eta^E) + R_2^E(\hat{\eta}^E_{j(i)},\eta^E)\} + o_P(n^{-1/2}).$$

Under the additional assumption 2, the central limit theorem yields $\sqrt{n}(\hat{\psi}^E - \psi^E) \rightsquigarrow N(0,\sigma^{2,E})$, where $\sigma^{2,E} = Var\{\varphi^E(X;\eta^E)\}$ is the non parametric efficiency bound in a model with observed data (W, A, Z, Z^π, M, Y).

- **Permutation**: Define $\mathbf{Z} = (Z_1, \ldots, Z_n) \in R^{n \times p}$ to be the data matrix, then we define $\mathbf{Z}^\pi = \mathbf{\Pi Z}$, with $\mathbf{Z}$ a doubly stochastic matrices with zero trace.
- **Approximation**: To approximate $\mathbf{Z}^\pi$ with $\mathbf{Z}$ with respect to $(\mathbf{A}, \mathbf{W})$, we want $(\mathbf{A}, \mathbf{W}) \simeq \mathbf{\Pi}(\mathbf{A}, \mathbf{W})$. Define a matrix $\mathbf{D}$ with $D_{i,j} = d((A_i, W_i), (A_j, W_j))$, then we solve

$$\min_{\mathbf{\Pi} \in \mathcal{P}} \sum_i d\{(A_i, W_i), (\mathbf{\Pi}(\mathbf{A}, \mathbf{W}))_i\} = \min_{\mathbf{\Pi} \in \mathcal{P}} \sum_{i,j} \Pi_{ij} D_{ij} = \min_{\mathbf{\Pi} \in \mathcal{P}} \mathrm{Tr}(\mathbf{\Pi D}),$$

and we use the simplex algorithm.